

Project Design Phase-II Data Flow Diagram & User Stories

Date	9 July 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)

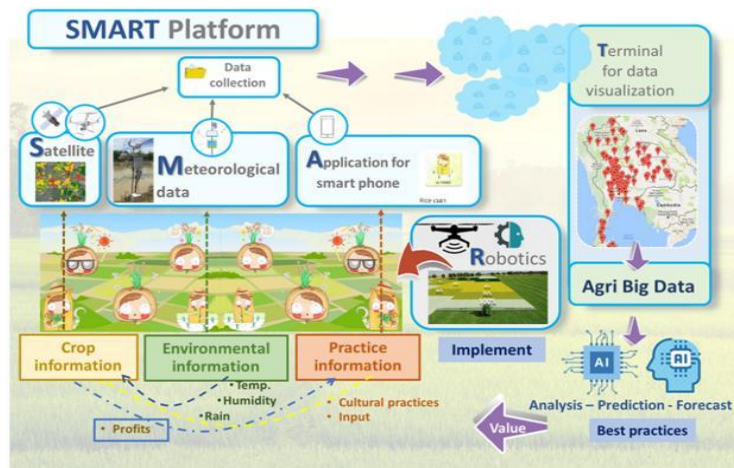
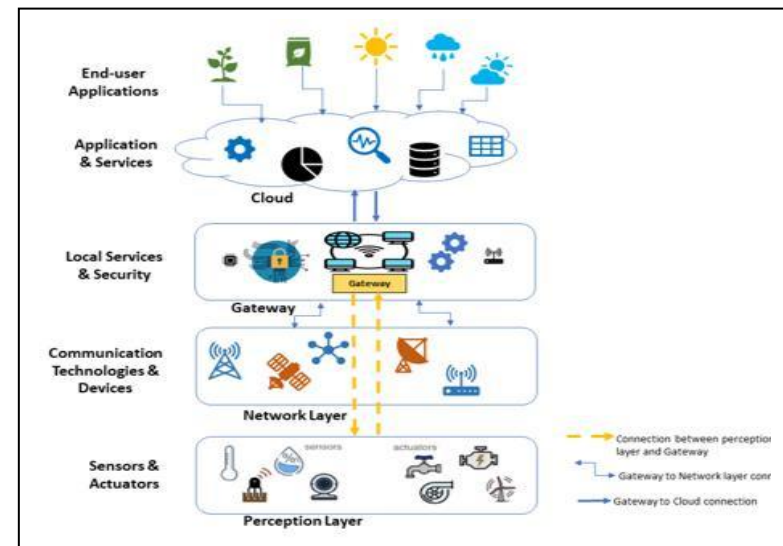


Figure 3. The schematic diagram of data flow in SMART platform



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
Farmer (Mobile User)	Login	USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
Farmer (Mobile User)	Crop Recommendation	USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2
Farmer (Mobile User)	Yield Prediction	USN-4	As a user, I can register for the application through Gmail	Predicted yield is shown	Medium	Sprint-1
Farmer (Mobile User)	Fertilizer Recommendation	USN-5	As a user, I can log into the application by entering email & password	Suitable fertilizers are displayed.	High	Sprint-1
Farmer (Mobile User)	Weather Monitoring	USN-6	As a farmer, I can view current weather and forecast information	Weather details are available on dashboard	Medium	Sprint-2
Farmer (Mobile User)	Dashboard	USN-7	As a farmer, I can view crop recommendations, yield predictions, and alerts in one dashboard	Dashboard displays all relevant information	High	Sprint-2
Customer Care Executive	Advisory Services	USN-8	As an expert, I can provide crop management advice to farmers	Farmers can view expert recommendations	Medium	Sprint-2
Administrator	User Management	USN-9	As an admin, I can manage farmer accounts	Admin can add, update, or delete users	High	Sprint-1
Administrator	Dataset Management	USN-10	As an admin, I can upload and update agricultural datasets	System uses updated datasets for predictions	High	Sprint-2
Administrator	Reports	USN-11	As an admin, I can generate reports on crop recommendations and yield predictions	Reports are generated successfully	Medium	Sprint-3
Administrator	System Monitoring	USN-12	As an admin, I can monitor system performance and user activity	Monitoring dashboard is available	Medium	Sprint-3